



# Measuring Compact Local Evidence Carriers

## A Reproducibility Audit of Qorx

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Artifact-backed research report

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### Abstract

Systems that keep evidence outside a language-model request can expose a short address while retaining a much larger local index. The address, the resolved evidence, and the eventual model input are different experimental objects. We define an accounting model that separates indexed corpus, carrier, resolved evidence, provider input, and task outcome, then apply it to three archived Qorx runs. In the headline artifact, five fixed queries each selected two local handles encoded in a 56-byte carrier. The artifact’s deterministic character-count estimator assigns 14 tokens to that carrier. Dividing an estimated 184,789,445 indexed tokens by fourteen gives an index-to-handle addressing ratio of 13,199,246.07x. Mean in-process lookup was 0.8974 ms; the median was 0.242 ms. The fixed two-handle carrier makes this a measurement of carrier construction and receipt integrity. Retrieval quality, semantic compression, and task performance require separate endpoints. Three archives share recorded binary and harness hashes and pass their declared gates, establishing byte-level repeatability of the recorded path. Source-to-binary reconstruction is the next reproducibility step and depends on recovering the historical source revision. A separate three-task Qorx 1.0.6 fixture records roughly 80x lower provider-reported input under fixed conditions. We present that observation with its one-call-per-task design and leave population inference to the confirmatory study. The audit closes with a protocol that joins retrieval quality, provider work, reproducibility, and environmental measurement.

**Evidence scope.** The May package supplies three integrity-checked archives, two reproduction scripts, and one companion record labeled cross-droplet reproduction. These materials support receipt verification, exact recomputation, and documented protocol reruns. The public repository does not identify the exact source revision corresponding to the recorded May binary and harness hashes, so independent source-to-binary reconstruction is a follow-on step. This author-issued preprint has not undergone journal peer review.

**Keywords:** evidence carrier; retrieval; context accounting; artifact evaluation; reproducibility; agent memory; local AI runtime

## 1 Introduction

Agent workflows repeatedly consult source files, logs, prior tool output, and task state. Putting all of that material into every model request is simple, but it spends context on information that may not be relevant to the current decision. Retrieval-augmented generation addresses a related problem by keeping knowledge outside model parameters and retrieving it when needed [6]. ReAct treats retrieval and other tools as actions within a reasoning loop [23]. Long-context studies also show that making a prompt larger does not guarantee that a model will use every relevant fact reliably [8].

Qorx implements this pattern as a local runtime. It indexes workspace material, returns a small carrier, and resolves the carrier when a task requires evidence. The carrier is an address, not the evidence itself. The distinction matters because a ratio between index size and address size can be numerically large even when retrieval quality and model-facing input remain unknown.

This report asks three questions:

1. What quantity does the archived 13,199,246.07x figure measure?
2. What level of repeatability and reproduction do the surviving artifacts support?
3. Which further measurements are required for performance, provider, and environmental conclusions?

The paper makes three contributions:

1. an accounting model that keeps index, carrier, resolved evidence, provider input, and task outcome separate;
2. a receipt-level audit of three May 2026 Qorx archives and two small public Qorx 1.0.6 checks; and
3. a confirmatory-study design that joins retrieval quality, end-to-end cost, and reproducibility instead of treating a token quotient as a system result.

The May experiment evaluates carrier construction. The confirmatory protocol extends that measurement to language-model behavior, coding-agent performance, security, long-horizon memory, and environmental impact.

## 2 Operational model

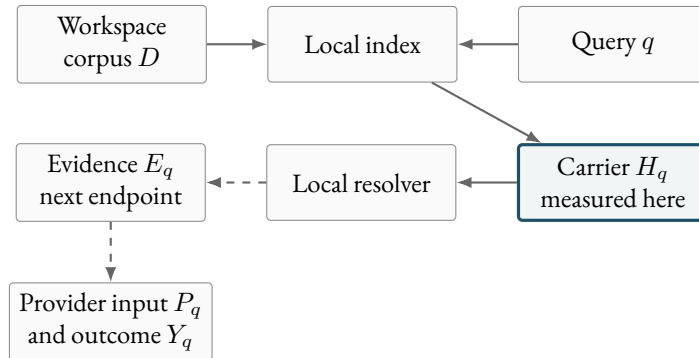
Let  $D$  be the local indexed corpus for a workspace,  $q$  a query,  $H_q$  the carrier returned for that query, and  $E_q$  any evidence later resolved from the carrier. Let  $P_q$  be the provider-bound input and  $Y_q$  the task outcome. Let  $\tau(x)$  denote a stated token estimator. Three ratios are useful when their numerators and denominators use compatible estimators:

$$\rho_H(q) = \frac{\tau(D)}{\tau(H_q)}, \quad \text{index-to-handle addressing ratio,} \quad (1)$$

$$\rho_E(q) = \frac{\tau(D)}{\tau(E_q)}, \quad \text{index-to-evidence payload ratio,} \quad (2)$$

$$\rho_P(q) = \frac{\tau(P_q^B)}{\tau(P_q^Q)}, \quad \text{baseline-to-Qorx provider-input ratio.} \quad (3)$$

The May benchmark reports  $\rho_H$ . Evaluating  $\rho_E$  and  $Y_q$  requires resolved excerpts, relevance judgments, and task outcomes. The separate July fixture supplies an initial  $\rho_P$  observation for three tasks; a larger randomized sample is needed for a population estimate.



**Figure 1:** Index, carrier, evidence, provider input, and task outcome are different experimental objects. The May microbenchmark measures the highlighted carrier path; dashed paths define the next evaluation endpoints.

### 2.1 Token estimator

The artifact uses the estimator

$$\tau_{\text{char4}}(x) = \max \left( 1, \left\lceil \frac{\text{characters}(x)}{4} \right\rceil \right).$$

#### Estimator boundary

The `char4` estimator is local character-count accounting. Provider tokenizer counts, semantic compression, and billable tokens are separate quantities and should be reported under their own estimators.

### 3 Measured implementation path

The archived method note identifies the measured path as a Quas search with QBH carrier accounting. “Proof tape” is the implementation name for the serialized record rather than a claim of formal proof semantics. The index contains path terms, symbols, signature hashes, and local links between referenced symbols. For each query, the benchmark selects at most two quarks. The QBH carrier stores an 8-byte frame header and a 24-byte handle for each selected quark:

$$\text{bytes}(H_q) = 8 + 24k.$$

With  $k = 2$ , the carrier is 56 bytes and  $\tau_{\text{char4}}(H_q) = 14$ . The carrier contains handles; a downstream model receives excerpt text only after resolution.

The experiment isolates Quas search and QBH accounting. A broader system study would add the other documented retrieval pipelines, Orcl traversal, prompt caching, the Qorx compiler, the virtual machine, QPie planning, and device leasing.

## 4 Artifact and method

### 4.1 Artifact identity

The audited archive is `qorx-context-reduction-x1-fever12500k-20260529T163955Z`. It was generated on May 29, 2026 UTC and described locally on May 30 in Asia/Manila. The archive declares an AMD MI300X x1 target and Qorx version `0.0.1-a-y1em`. The hardware label records provenance. Because the receipt omits accelerator utilization, the search span is reported as an in-process runtime measurement rather than a GPU performance result.

| Field               | Recorded value                                                           |
|---------------------|--------------------------------------------------------------------------|
| Archive SHA-256     | 09dd67e3e15913cb1e71f575f525014a3184e73ebf58e767a400fc933d7c2332         |
| Qorx binary SHA-256 | 42be06b3991c646aa42eed8bf17b76ba2afa067638123309fa41a599ebfa15c          |
| Harness SHA-256     | 6e6a64557c4ab34962a5d144ba6efa2aea2120f09ad2e265e771fc2ad19f1d23         |
| Measurement mode    | <code>in_process_zero_text_proof_tape_excludes_cli_and_index_load</code> |

**Table 1:** Identity fields copied from the archived receipt. The companion manifest records the local paths and verification result.

The corpus combined Qorx source with public BEIR SciFact, NFCorpus, ArguAna, FiQA, and FEVER records. The receipt reports 2,527 files and 1,006 MB of source data. It assigns 184,789,445 estimated tokens to the index. The archive carries manifests and a reconstruction script rather than the full corpus and source snapshot. The script redownloads the public BEIR distributions, repeats the recorded splitting procedure, copies a Qorx source tree from `/root/qorx`, and invokes a Qorx executable at a declared path. Exact reconstruction therefore depends on the source tree and executable identified by the recorded hashes.

### 4.2 Companion May runs and reproducibility level

The evidence directory contains three May archives. Their outer SHA-256 sidecars match, and every archive contains thirteen member digests that were reverified for this revision. All three record Qorx binary SHA-256 `42be06b3...15c` and harness SHA-256 `6e6a6455...1d23`, exit with status zero, and pass their declared gates. The smaller no-FEVER package is labeled a cross-droplet reproduction in its companion README. Because the archive does not encode host identity, this report uses “cross-droplet” as the recorded descriptor.

This package establishes four reproducibility levels: the receipts are integrity-verifiable; the recorded arithmetic is exactly recomputable; the benchmark procedure is rerunnable when the identified source tree and executable are present; and a same-author run labeled cross-droplet reproduction is documented. Independent source-to-binary rebuilding is the remaining level and requires the exact hashed source revision, dependency environment, and compiler attestation.

| May archive                     | Files | Indexed estimate | Addressing ratio | Mean / max ms  |
|---------------------------------|-------|------------------|------------------|----------------|
| No-FEVER, labeled cross-droplet | 396   | 17,605,091       | 1,257,506.50x    | 0.6314 / 2.207 |
| FEVER 10M gate                  | 2,045 | 147,983,583      | 10,570,255.93x   | 0.7206 / 2.626 |
| FEVER 12.5M headline            | 2,527 | 184,789,445      | 13,199,246.07x   | 0.8974 / 3.512 |

**Table 2:** Three integrity-checked May runs under the same recorded binary and harness hashes. The fixed 14-token carrier makes the ratio scale with the index estimate and provides repeatability evidence for the benchmark path.

### 4.3 Queries and measurements

The harness executed five fixed queries. Each row selected two quarks and produced the same 56-byte tape.

| ID | Query                                               | Quarks | Tape bytes | Core ms |
|----|-----------------------------------------------------|--------|------------|---------|
| Q1 | qorx codex hooks quark pointer mcp                  | 2      | 56         | 3.512   |
| Q2 | context fault proof pages resolver boundary         | 2      | 56         | 0.396   |
| Q3 | strict answer refusal unsupported claims            | 2      | 56         | 0.208   |
| Q4 | qlp quantum leap qubo local quarks                  | 2      | 56         | 0.242   |
| Q5 | benchmark reproduction external artifact boundaries | 2      | 56         | 0.129   |

**Table 3:** All query rows in the archived JSON. Every observation, including the slower first query, is retained; the fixed order leaves a cold-versus-warm interpretation unresolved.

| Quantity                         | Value                      |
|----------------------------------|----------------------------|
| Queries                          | 5                          |
| Indexed-token estimate           | 184,789,445                |
| Carrier size                     | 56 bytes / 14 char4 tokens |
| Index-to-handle addressing ratio | 13,199,246.07x             |
| Mean core lookup latency         | 0.8974 ms                  |
| Median core lookup latency       | 0.242 ms                   |
| Minimum / maximum core latency   | 0.129 / 3.512 ms           |
| Quas build latency               | 12,239.854 ms              |
| Recorded Quas wall latency       | 18,049.631 ms              |
| Full suite elapsed time          | 175.325 s                  |
| Warm-up queries excluded         | 0                          |
| Provider-call telemetry          | Unobserved                 |

**Table 4:** Descriptive results recomputed from the raw JSON. The mean and maximum match the receipt; the median and minimum are derived from the five rows.

The addressing ratio follows directly from the two recorded estimates:

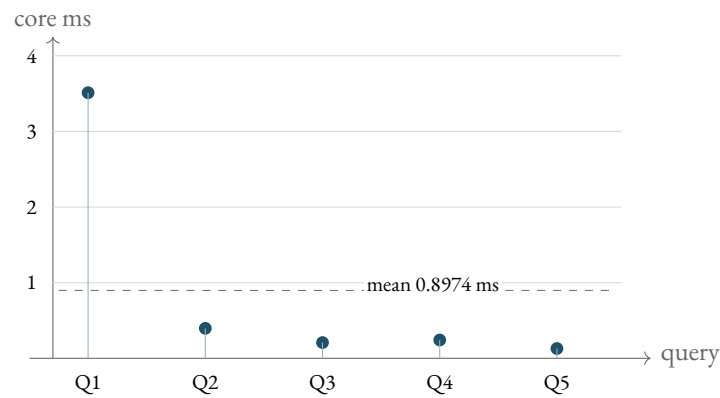
$$\rho_H = \frac{184,789,445}{14} = 13,199,246.0714 \dots$$

## 5 Results and interpretation

The primary result is an index-to-handle addressing calculation. The fourteen-token value describes a fixed local address for two selected handles; it should be read separately from semantic compression of the 184,789,445-token index estimate.

The experiment shows that this implementation can serialize two selected quark references into a fixed 56-byte local carrier and perform the recorded in-process lookup. The ratio quantifies addressability: it compares a large index estimate with the fixed-size address. Measuring semantic compression would also require the resolved evidence and downstream task result.

The carrier stayed at fourteen estimated tokens in all three May archives, which isolates the indexed-corpus estimate as the source of the increasing ratios. A variable-carrier experiment is needed to model representation cost across query complexity, while resolved excerpts and relevance judgments would enable evidence-sufficiency analysis.



**Figure 2:** All five core lookup latencies. Q1 accounts for most of the difference between the mean and median, and every observation is retained.

Q1 took 3.512 ms, while the remaining four rows ranged from 0.129 to 0.396 ms. The fixed order and single pass leave warm-up, cache state, query difficulty, and noise unresolved as explanations for the difference. The receipt also records a 12,239.854 ms build field, an 18,049.631 ms wall field, and a 175.325 s suite duration. These are separate timing boundaries; system-level reporting should retain all of them.

The artifact identifies the measured path as in-process and local. Its JSON scope ends before provider activity and omits an explicit provider-call counter, so provider work is unobserved rather than recorded as zero. Provider cost and privacy belong in an instrumented end-to-end workflow.

## 6 Exploratory public Qorx 1.0.6 checks

Two July 22, 2026 artifacts exercise versioned public Qorx 1.0.6 snapshots. They form a separate evidence layer from the May implementation and are reported independently of the 13.2-million addressing ratio.

The repository check indexed 407,902 local `char4` tokens. A session handle exposed 69 estimated tokens for a 5,911.62x local accounting ratio; a query pack exposed 410 tokens for 994.88x. One supported strict task and one deliberately unsupported task both met their expected outcomes, and the local agent path recorded zero provider calls. These deterministic results verify the disclosed repository fixture. A general retrieval evaluation requires a larger held-out task set. The commands and raw output are public.<sup>1</sup>

The DataCamp check compared full-context prompts with Qorx evidence packs for two exact-answer questions and one unsupported question. It used the Qorx 1.0.6 static Linux release, a fixed 106,758-character synthetic corpus, temperature zero, an 80-token output cap, and one call per question and condition.

| Provider adapter                          | Full input | Qorx input | Ratio  | Full tasks | Qorx tasks |
|-------------------------------------------|------------|------------|--------|------------|------------|
| OpenAI, <code>gpt-4o-mini</code>          | 65,850     | 814        | 80.90x | 3/3        | 3/3        |
| Anthropic, <code>claude-sonnet-4-6</code> | 77,458     | 963        | 80.43x | 3/3        | 3/3        |

**Table 5:** Aggregate provider-reported input tokens and deterministic fixture outcomes. Uncached billing, latency, quality, and environmental performance require separate measurements.

The full-context condition always ran first. Two OpenAI full-context calls reported 21,888 cached input tokens each; the Anthropic rows reported none. That ordering and unequal cache reporting leave latency uncontrolled. The 3/3 fixture result is descriptive because the design has one call per task and condition. The normalized capture, questions, hashes, answers, cache fields, and reproduction notebook are public.<sup>2</sup>

### 6.1 Post-freeze implementation status

The quantitative evidence in this report was frozen at the July 22 public artifacts. A local source snapshot inspected on July 31 adds IDRA and draft auto-index, persistent-index, QVP, and response-reuse files. Its compiled `qorx`

<sup>1</sup>Repository artifact at commit `fd7589`.

<sup>2</sup>Provider artifact at commit `362e142`.

target passed 363 tests, and all targets passed cargo check. The inspection also found a formatting failure, four feature files outside the crate module graph, an uncommitted source tree, and a source-to-installed-binary provenance gap. These findings describe implementation status rather than experimental performance. The supplementary audit records the details, while the paper’s quantitative results remain tied to the May archives and public Qorx 1.0.6 fixtures.

## 7 Connecting token accounting to environmental measurement

Longer inputs can increase prefill work, while production serving also depends on output length, batching, hardware, prefix reuse, and key–value cache state [17, 2, 16]. A smaller provider input may change energy use under some configurations. Quantifying that change requires workload and facility measurements alongside token accounting.

The available receipts contain token and latency fields while omitting power, facility, electricity-mix, cooling, water, and embodied-hardware telemetry. They also lack the matched end-to-end baseline needed to allocate local indexing and resolution overhead. Environmental outcomes therefore remain an open measurement target rather than a conversion from the token quotient.

The supplementary audit specifies a paired protocol with fixed task quality, randomized condition order, repeated runs, cache-state reporting, index amortization, and separate operational-energy, carbon, water-consumption, and water-withdrawal boundaries [4, 7, 15]. Under that protocol, environmental benefit becomes a testable outcome.

## 8 Measurement boundaries and next tests

**Measurement target.** The dependent variable is handle size. Semantic compression, retrieval quality, and answer quality are separate endpoints in the confirmatory protocol. “Context reduction” is retained only as the historical name used by the artifact.

**Estimator.** Both sides of the ratio use local estimates. The 14-token denominator is  $\lceil 56/4 \rceil$ ; a named model tokenizer would produce its own denominator and ratio.

**Sample size.** The five developer-written queries support row-level description rather than population inference. The report therefore publishes every row and limits its statistics to the mean, median, and range.

**Statistical reporting.** The May and July observations use descriptive accounting ratios and observed latencies under fixed corpus, query, carrier, and ordering conditions. All recorded rows are retained. A confirmatory sample should preregister its power analysis, uncertainty intervals, distributional assumptions, and multiplicity handling instead of applying significance language to this small audit [19].

**Timing boundary.** Core lookup excludes process startup and index loading. The first query is slower than the other four, and every observation is retained. End-to-end latency should combine the separately recorded build, wall, and suite boundaries under a repeated design.

**Reproducibility level.** The May directory provides three verified archives, two shell scripts that reconstruct the public dataset slices and invoke the benchmark, and a same-author run labeled cross-droplet reproduction. These materials support receipt verification, exact arithmetic recomputation, and protocol-level rerunning when the identified Qorx source tree and executable are supplied. Independent source-to-binary reproduction is the next level and requires the exact historical source revision linked to the recorded binary and harness hashes.

**Generalization.** The run uses one archived corpus, one declared hardware target, one fixed limit, and one local estimator. Confirmatory runs should vary repositories, hardware, query sets, carrier limits, and agent tasks while recording accelerator utilization.

**Environmental scope.** The receipt contains latency and local accounting fields. A matched environmental study would add power traces, facility telemetry, electricity mix, cooling data, water factors, embodied-hardware allocation, and an end-to-end baseline.

**Public-fixture scope.** The Qorx 1.0.6 provider check contains three synthetic tasks, one call per condition, fixed order, heterogeneous provider caching, and deterministic substring scoring. It verifies the disclosed fixture and motivates a larger randomized provider study.



## 9 Related work

Qorx sits between retrieval, prompt compression, and external agent memory. This paper contributes a measurement boundary for compact carriers; the surrounding fields supply the task-level benchmarks needed to evaluate the broader system.

**Retrieval and compression.** RAG retrieves external passages for generation [6], while GraphRAG organizes evidence through graph-structured summaries [3]. LLMingua and LLMingua-2 shorten readable prompts and test the compressed prompts on downstream tasks [5, 12]. RECOMP compresses retrieved documents and can omit irrelevant augmentation [22]. The QBH object instead addresses evidence through a local resolver. Its natural evaluation pairs carrier accounting with resolved-evidence and downstream-task measurements.

**External memory and compact evidence.** MemGPT manages context through memory tiers [11]. LoCoMo and LongMemEval test conversational memory across sessions, including temporal reasoning and abstention [9, 21]. LongMemEval-V2 frames memory as context gathering from environment histories and evaluates the compact evidence returned for downstream questions [20]. AMA-Bench evaluates memory over agent-environment trajectories [24]. These benchmarks define natural task-level extensions to the five-query carrier audit.

**Evaluation and reproducibility.** TREC and BEIR establish comparative retrieval evaluation with shared queries, relevance judgments, and strong lexical baselines [10, 18]. RAGChecker and SURE-RAG separate retrieval, evidence sufficiency, answer quality, and abstention [14, 13]. ACM artifact terminology separates availability, functional evaluation, repeatability, and reproduction [1]. Within that literature, this paper contributes an accounting and reproducibility audit of one compact carrier implementation.

## 10 Confirmatory study design

A confirmatory Qorx study should:

1. freeze a public source revision, dependency lock, compiler, build command, corpus manifest, and executable digest before measurement;
2. use held-out tasks drawn from established retrieval and memory benchmarks, together with repository tasks that have executable oracles;
3. compare full context, BM25, frozen dense retrieval, hybrid retrieval, a readable compression baseline such as RECOMP or LLMingua, and Qorx under identical model and budget settings;
4. report carrier bytes, resolved evidence bytes, provider input and output, retrieval recall, ranking quality, evidence sufficiency, task success, abstention, latency, retries, and failures;
5. randomize condition order, repeat runs, publish uncertainty intervals, and use independent query authors or blinded evaluators; and
6. measure end-to-end energy only at matched task quality, with cache state, local indexing, provider work, and index amortization inside the declared boundary.

A privacy or security evaluation needs a threat model, controlled leakage study, and explicit observation of provider traffic. Locality and provider-call counts then become measured inputs to that analysis.

## 11 Ethics and disclosure

Local indexing can expose private source material if a resolver selects more text than a task requires. A small handle reduces what is present in the carrier, yet later resolution can still disclose sensitive text. Operators need explicit path controls, secret filtering, audit logs, and a way to disable capture.

Environmental reports should state their functional unit, quality threshold, hardware and provider boundary, cache policy, location, energy source, water boundary, uncertainty, and exclusions. Workload-specific measurements can then connect token accounting to energy, emissions, and water outcomes.

This paper was revised with language-model assistance for file inspection, calculation checks, editing, and LaTeX production. The author remains responsible for the claims. The companion evidence manifest identifies each quantitative statement and its source receipt.

**Competing interests and governance.** The author created Qorx and is affiliated with IDRA, creating intellectual property and reputational interests in the system. The study is author-led without an independent sponsor or artifact evaluator. Private materials were used only to delimit development status; every quantitative claim in the paper is tied to the listed May archives or public July artifacts. Any financial or commercial interests existing at submission should also be entered in the target venue’s required form.

**Dataset permissions.** The May scripts download publicly distributed BEIR packages whose licenses and notices vary by collection. Any archival deposit or redistribution must preserve those dataset-specific terms and citation requirements.

## 12 Conclusion

Across three May archives, Qorx repeatedly produced a 56-byte address for two selected local records under the same recorded binary and harness hashes. In the headline run, dividing the 184,789,445-token index estimate by the fourteen-token carrier estimate gives 13,199,246.07x. The result is an index-to-handle addressing ratio, and both the receipts and arithmetic are verifiable.

The audit also identifies the measurements needed to move from carrier accounting to system evaluation: retrieval relevance, resolved evidence, answer quality, end-to-end latency, provider work, and environmental telemetry. Independent source-to-binary reconstruction requires the exact historical source revision as well. The separate July fixture contributes an early provider-input observation of roughly 80x on three synthetic tasks; its one-call design motivates a larger randomized study.

The resulting reporting rule is simple: report index size, carrier size, resolved evidence, provider input, and task outcome separately. This audit uses that rule to specify a confirmatory test of Qorx on the quality–cost frontier.

## Artifact availability

The public documentation surface and versioned Qorx 1.0.6 snapshots are available at <https://github.com/bbrainfuckk/qorx>. The accompanying evidence package contains three archived run bundles, matching outer sidecars, internally verified member digests, two rerun scripts, method notes, a claim register, and supplementary audit material. Table 2 records the supported reproducibility levels and the source-to-binary step that remains. The measured May implementation and the later local development snapshot are separate from the public 1.0.6 source. A stable public deposit identifier will be added after licensing review and archival deposit.

## References

- [1] Association for Computing Machinery. Data and software artifacts, 2026. Accessed 2026-07-23.
- [2] Hiari Pizzini Cavagna, Andrea Proia, Giacomo Madella, Giovanni B. Esposito, Francesco Antici, Daniele Cesarini, Zeynep Kiziltan, and Andrea Bartolini. SweetSpot: An analytical model for predicting energy efficiency of LLM inference. In *Proceedings of the 17th ACM/SPEC International Conference on Performance Engineering*, 2026.
- [3] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization, 2024.
- [4] Green Software Foundation. Software carbon intensity specification, version 1.1.0. Technical report, Green Software Foundation, 2024.
- [5] Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. LLMingua: Compressing prompts for accelerated inference of large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13358–13376, 2023.
- [6] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- [7] Pengfei Li, Jianyi Yang, Mohammad A. Islam, and Shaolei Ren. Making AI less “thirsty”: Uncovering and addressing the secret water footprint of AI models. *Communications of the ACM*, 68(7):54–61, 2025.
- [8] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.



- [9] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 13851–13870, 2024.
- [10] National Institute of Standards and Technology. About the text REtrieval conference (TREC). NIST Information Technology Laboratory, 2026. Accessed 2026-07-23.
- [11] Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems, 2023.
- [12] Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Ruhle, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Dongmei Zhang. LLMingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.
- [13] Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation, 2026.
- [14] Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Chengwei Wang, Shichao Sun, Huanyu Li, Zizhao Pandey, Sirui Wang, Tuan Dung Nguyen, Edward Y. Ma, Qian Peng, Zhengzhong Zeng, Chun-Hao Yu, Hanieh Deilamsalehy Chawla, Zhenchang Xu, Bryan Hsieh, Sunil Nayak, Timmy Zhang, Han Wang, Caiming Cheng, Yangqiu Yin, and Heng Ji. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation, 2024.
- [15] Arman Shehabi, Sarah J. Smith, Alex Hubbard, Alex Newkirk, Nuoa Lei, Md Abu Bakar Siddik, Billie Holecek, Jonathan Koomey, Eric Masanet, and Dale Sartor. 2024 united states data center energy usage report. Technical Report LBNL-2001637, Lawrence Berkeley National Laboratory, 2024.
- [16] Lola Solovyeva and Fernando Castor. Towards green AI: Decoding the energy of LLM inference in software development, 2026.
- [17] Vikranth Srivatsa, Zijian He, Reyna Abhyankar, Dongming Li, and Yiyang Zhang. Preble: Efficient distributed prompt scheduling for LLM serving, 2024.
- [18] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.
- [19] Ronald L. Wasserstein and Nicole A. Lazar. The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2):129–133, 2016.
- [20] Di Wu, Zixiang Ji, Asmi Kawatkar, Bryan Kwan, Jia-Chen Gu, Nanyun Peng, and Kai-Wei Chang. LongMemEval-V2: Evaluating long-term agent memory toward experienced colleagues, 2026.
- [21] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025.
- [22] Fangyuan Xu, Weijia Shi, and Eunsol Choi. RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation. In *International Conference on Learning Representations*, 2024.
- [23] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.
- [24] Yujie Zhao, Boqin Yuan, Junbo Huang, Haocheng Yuan, Zhongming Yu, Haozhou Xu, Lanxiang Hu, Abhilash Shankarampeta, Zimeng Huang, Wentao Ni, Yuandong Tian, and Jishen Zhao. AMA-Bench: Evaluating long-horizon memory for agentic applications, 2026.